Tyler Weber

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LinkedIn | Github | Personal Portfolio

EDUCATION:

University of North Carolina at Charlotte, Charlotte, NC **May 2026**
B.S. in Computer Science, Concentration in Data Science **GPA: 3.96**

- Honors: **Chancellor's List** (Fall 2023 - Spring 2026)
- **Relevant Coursework:** Data Mining, Visual Analytics, Cloud Computing, Machine Learning, Data Structures & Algorithms, Database Design & Implementation, Backend Application Development, Software Engineering

TECHNICAL SKILLS:

Languages: Python, SQL, Java, HTML, CSS
Tools & Libraries: Pandas, Matplotlib, Seaborn, Scikit-Learn, Jupyter Notebooks, Excel, Power BI, Tableau, Cursor, Claude
Technologies: PostgreSQL, Prisma, FastAPI, REST APIs, OpenAPI, Swagger UI, CORS, Git, Linux, Visual Studio, JUnit Testing
Core Foundations: Data Cleaning, Visualization, Machine Learning, Backend Development, Web Development

WORK EXPERIENCE

Data Analyst Intern | Technosylva *February 2026 - Present*

- Collaborate with a team of data scientists on weather-related power outage risk models using large-scale weather and outage datasets.
- Developed an internal ML model validation dashboard for monitoring model health, prediction accuracy, and geographic outage predictions in real time.
- Support performance tracking features for RMSE, MAE, MAPE, rolling error analysis, and model drift detection across outage prediction models.

PROJECTS:

FlashCut AI Flashcard Generator | GitHub *April 2026*

Developed a full-stack Chrome extension that converts highlighted web content into AI-generated flashcards.

- Collaborated with a 5-person team to build a Chrome extension that captures selected webpage text and generates context-aware flashcards.
- Implemented backend functionality using Node.js, Express, and PostgreSQL to validate requests, process AI-generated flashcards, and persist user data.
- Built REST API functionality for authentication, flashcard generation, flashcard retrieval, and extension-to-backend communication.
- Supported secure, maintainable architecture using JWT-protected routes, password hashing, environment variables, controllers, services, and database models.

Madden NFL Position Classification | Portfolio *October 2025*

Developed a machine learning model to predict NFL player positions using Madden ratings and physical attributes.

- Achieved 94% test accuracy and 91% macro F1 score by engineering key features and optimizing the model.
- Performed data preprocessing, including cleaning and normalization, to set up and improve model performance.
- Built with Python, scikit-learn, pandas, and visualization libraries, producing insightful visualizations and actionable insights.

STUDENT INVOLVEMENT

Computing & Informatics Learning Community - UNCC *August 2023 - May 2024*

- One of 48 students selected for a cohort focused on technical collaboration, networking, and project-based learning in computer science.
- Participated in coding workshops and mentorship sessions emphasizing soft skills and professional growth.